

A dissipation-based arc-length method for robust simulation of brittle and ductile failure[‡]

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SUMMARY

A robust method to trace the equilibrium path in non-linear solid mechanics problems is proposed. A general arc-length constraint based on the energy release rate is developed. Constraints have been derived for the cases of geometrically linear damage, geometrically linear plasticity and geometrically non-linear damage. All three constraints can efficiently be applied in a finite element context. Numerical simulations demonstrate that the proposed framework gives robust results for these cases. Applicability of the proposed framework to other types of constitutive and/or kinematic behaviour is predicted. Copyright © 2008 John Wiley & Sons, Ltd.

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1. INTRODUCTION

Computations for quasi-static non-linear solid mechanics problems are often carried out for the purpose of determining important characteristics of the body under consideration. Examples of such characteristics are the maximum load or the residual strength at a given strain level. Determination of such properties often requires tracing of the whole equilibrium path. The availability of a robust method for stepwise determination of the required points of the equilibrium path is therefore indispensable.

Several approaches have been proposed in the past. The pioneering work of Riks [1] is worthy of mention, together with the alternative formulations by Ramm [2] and Crisfield [3]. A comprehensive

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‡This paper is in honour of the achievements of René de Borst on the occasion of his 50th birthday.

review of the available techniques is provided by Geers [4, 5]. The traditional approach consists of parametrizing the equilibrium path with its own arc parameter, i.e. with the norm of the incremental degree-of-freedom vector, which also motivates the generic ‘arc-length’ denomination for this kind of method. The arc parameter works properly for problems exhibiting geometrical non-linearities but often fails when material instabilities are involved that would lead to localized failure process zones. A remedy for this problem is to consider only the degrees of freedom involved in the failure process to be coupled to the path parameter [6]. Although such an approach is robust in general, it is not applicable when the location or behaviour of the failure process zone is not *a priori* predictable. This is the case when Monte Carlo simulations of failure of heterogeneous materials are carried out or when crack propagation or interfacial delamination is the relevant dissipative phenomenon. Remedies for this problem have been proposed in [4, 5, 7], in which the path parameter is coupled to smartly selected internal variables.

A path-following constraint based on the energy release rate in the case of a geometrically linear continuum damage model was introduced in [8]. This method has the advantage that the dissipated energy is a global quantity and therefore no *a priori* selection of degrees of freedom is required. Moreover, as such a constraint is directly related to the failure process itself, a stable convergence behaviour is observed even for far advanced stages of the equilibrium path.

In this contribution the path-following constraint as proposed in [8] will be extended to the case of geometrically linear plasticity computations and geometrically non-linear damage computations. In addition, the method will be applied in the context of the partition of unity method.

2. ENERGY RELEASE RATE PATH-FOLLOWING CONSTRAINT

Consider the body Ω as shown in Figure 1 with boundary conditions

$$\mathbf{n}_t \cdot \boldsymbol{\sigma} = \bar{\mathbf{t}}, \quad \mathbf{x} \in \Gamma_t \quad (1)$$

$$\mathbf{u} = \bar{\mathbf{u}}, \quad \mathbf{x} \in \Gamma_u \quad (2)$$

where $\boldsymbol{\sigma}$ is the Cauchy stress in the material, $\bar{\mathbf{t}}$ are the prescribed tractions on the external boundary Γ_t with outward normal vector $\mathbf{n}_t$ and $\bar{\mathbf{u}}$ are the prescribed displacements on Γ_u . The principle of virtual work is applied to yield the quasi-static equilibrium of the body as

$$\int_{\Omega} \boldsymbol{\sigma}^T \delta \boldsymbol{\varepsilon} d\Omega = \int_{\Gamma_t} \bar{\mathbf{t}}^T \delta \mathbf{u} d\Gamma \quad (3)$$

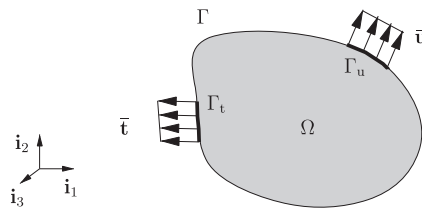


Figure 1. A body Ω with boundary Γ subjected to external loads $\bar{\mathbf{t}}$ and prescribed displacements $\bar{\mathbf{u}}$.

where body forces have been neglected and where $\delta \mathbf{u}$ and $\delta \boldsymbol{\varepsilon}$ are the admissible displacement and engineering strain field, respectively. The displacement field $\mathbf{u}(\mathbf{x})$ can be written in discrete form using standard finite element shape functions and nodal displacements $\mathbf{a}$ by

$$\mathbf{u} = \mathbf{N}\mathbf{a} \quad (4)$$

where matrix $\mathbf{N}$ contains the finite element shape functions. The admissible displacement field $\delta \mathbf{u}$ is then

$$\delta \mathbf{u} = \mathbf{N}\delta \mathbf{a} \quad (5)$$

Finally, the admissible strain in discrete form can be formulated as

$$\delta \boldsymbol{\varepsilon} = \mathbf{B}\delta \mathbf{a} \quad (6)$$

where matrix $\mathbf{B}$ maps the nodal displacement to the virtual strain field. Substituting the discrete relations into the equilibrium equation (3) yields the system of N equations

$$\int_{\Omega} \mathbf{B}^T \boldsymbol{\sigma} d\Omega = \int_{\Gamma_t} \mathbf{N}^T \bar{\mathbf{t}} d\Gamma \quad (7)$$

The left-hand side of this equation represents the internal force $\mathbf{f}_{\text{int}}$, which is a function of the displacement field $\mathbf{a}$. The right-hand side represents the external force vector $\mathbf{f}_{\text{ext}}$, which is generally independent of the displacement field $\mathbf{a}$ such that

$$\mathbf{f}_{\text{int}}(\mathbf{a}) = \mathbf{f}_{\text{ext}} \quad (8)$$

The external forces can be expressed as a unit force vector $\hat{\mathbf{f}}$ multiplied with a load factor λ to yield

$$\mathbf{f}_{\text{int}}(\mathbf{a}) = \lambda \hat{\mathbf{f}} \quad (9)$$

For each value of the load factor λ , a solution $\mathbf{a}$ of this system of equations can be obtained. The collection of equilibrium points $(\mathbf{a}, \lambda)$ of the system is referred to as the equilibrium path. In practice, this equilibrium path is traced in an incremental fashion. Given a point on the equilibrium path $(\mathbf{a}_0, \lambda_0)$, the next point on the path can be computed by solving the set of non-linear equations

$$\mathbf{f}_{\text{int}}(\mathbf{a}_0 + \Delta \mathbf{a}) = [\lambda_0 + \Delta \lambda] \hat{\mathbf{f}} \quad (10)$$

for the incremental displacement $\Delta \mathbf{a}$ and incremental load factor $\Delta \lambda$. Since this system of N equations has $N + 1$ unknowns (the degrees of freedoms $\mathbf{a}$ plus the load factor λ), it is indeterminate. In order to solve it, an additional constraint equation needs to be specified. Since this constraint prescribes the steps made on the equilibrium path, it is normally referred to as the path-following constraint. In general, this path-following constraint can be written as

$$g(\mathbf{a}_0, \lambda_0, \Delta \mathbf{a}, \Delta \lambda, \tau) = 0 \quad (11)$$

where τ is the prescribed path-parameter that determines the size of a step. The equilibrium state of this well-posed, augmented system of $N + 1$ equations can be solved simultaneously from

$$\begin{bmatrix} \mathbf{f}_{\text{int}} \\ g \end{bmatrix} = \begin{bmatrix} \lambda \hat{\mathbf{f}} \\ 0 \end{bmatrix} \quad (12)$$

This can be done in an iterative fashion by using a Newton–Raphson scheme. The solution $(\mathbf{a}, \lambda)$ at iteration number $k+1$ is equal to

$$\begin{bmatrix} \Delta \mathbf{a}^{k+1} \\ \Delta \lambda^{k+1} \end{bmatrix} = \begin{bmatrix} \Delta \mathbf{a}^k \\ \Delta \lambda^k \end{bmatrix} + \begin{bmatrix} \mathbf{K} & -\hat{\mathbf{f}} \\ \mathbf{h}^T & w \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{r}^k \\ -g^k \end{bmatrix} \quad (13)$$

where $\mathbf{K}$ and $\mathbf{r}$ are the stiffness matrix and the residual, which are, respectively, defined as

$$\mathbf{K} = \frac{\partial \mathbf{f}_{\text{int}}}{\partial \mathbf{a}}, \quad \mathbf{r} = \lambda \hat{\mathbf{f}} - \mathbf{f}_{\text{int}} \quad (14)$$

The vector $\mathbf{h}$ and the scalar w in Equation (13) are defined as

$$\mathbf{h} = \frac{\partial g}{\partial \mathbf{a}}, \quad w = \frac{\partial g}{\partial \lambda} \quad (15)$$

2.1. Energy release rate

The only requirement to the choice of constraint equations is that for a given set of unknowns $(\Delta \mathbf{a}, \Delta \lambda)$ the path parameter is monotonically strictly increasing. For the simulation of damage evolution, a natural choice is to use a constraint based on the rate of energy dissipation [8]. By virtue of the second law of thermodynamics, the rate of dissipation is non-negative. In the case of evolving damage, the rate of dissipation is positive and hence it is suitable to be used as a path parameter. Alternatively, when damage is not evolving, the rate of dissipation is equal to zero and it cannot be used to trace the equilibrium path. Path-following constraints that can trace such non-dissipative equilibrium paths (or parts thereof) are well developed. This contribution focuses on the development of a dissipation-based path-following constraint. The treatment of non-dissipative parts of the equilibrium path is further addressed in Section 3.3 and is illustrated with an example in Section 4.1.

The rate of dissipation of a body G is equal to the exerted power P minus the rate of elastic energy $\dot{V}$

$$G = P - \dot{V} \quad (16)$$

where $(\dot{})$ denotes the derivative with respect to time. In order to use the rate of dissipation as a path-following constraint, it should be expressed in terms of the nodal displacements $\mathbf{a}$, the load factor λ and the unit external force vector.

The exerted power is defined as the applied external force times the nodal velocity. In terms of the discretized model, this can be written as

$$P = \mathbf{f}_{\text{ext}}^T \dot{\mathbf{a}} = \lambda \hat{\mathbf{f}}^T \dot{\mathbf{a}} \quad (17)$$

The expression for the elastic energy stored in the solid depends on the constitutive behaviour of the material as well as on the kinematic formulation that is used. The rate of elastic energy and the rate of dissipation will be derived for three cases: (i) a geometrically linear kinematic formulation with damage, (ii) a geometrically linear kinematic formulation in combination with plasticity and (iii) a geometrically non-linear kinematic formulation in combination with damage. A path-following constraint will be derived for these three cases. Attention will also be paid to the formulation of the constraints in terms of a volume integral, which is useful in cases where

the rate of dissipation cannot be expressed in terms of the load factor and nodal displacements, which is the case in geometrically non-linear formulations in combination with plasticity.

2.1.1. Geometrically linear model and damage. In [8] a path-following constraint based on the energy release rate is proposed for a damage model in combination with a geometrically linear kinematic formulation. An important assumption for such a model is that the unloading behaviour is linear elastic and hence unloading occurs along the secant, as shown in Figure 2. The elastic energy stored in the solid can then be written as

$$V = \frac{1}{2} \int_{\Omega} \boldsymbol{\varepsilon}^T \boldsymbol{\sigma} d\Omega \quad (18)$$

where $\boldsymbol{\varepsilon}$ is the engineering strain and $\boldsymbol{\sigma}$ is the Cauchy stress. Assuming a small strain kinematic relation, the strain field can be expressed in terms of the nodal displacements as

$$\boldsymbol{\varepsilon} = \mathbf{B}\mathbf{a} \quad (19)$$

where the strain–nodal displacement matrix $\mathbf{B}$ is independent of the nodal deformation $\mathbf{a}$. Substituting this expression into (18) in combination with the expression for the internal force vector (9), the elastic energy is written as

$$V = \frac{1}{2} \mathbf{a}^T \int_{\Omega} \mathbf{B}^T \boldsymbol{\sigma} d\Omega = \frac{1}{2} \mathbf{a}^T \mathbf{f}_{\text{int}} \quad (20)$$

Assuming that the system is in equilibrium, relation (8) holds and the elastic energy can be written in terms of the nodal displacement and the force vector only as

$$V = \frac{1}{2} \lambda \mathbf{a}^T \hat{\mathbf{f}} \quad (21)$$

Taking the derivative of this equation with respect to time, the rate of change of the elastic energy equals

$$\dot{V} = \frac{1}{2} \dot{\lambda} \mathbf{a}^T \hat{\mathbf{f}} + \frac{1}{2} \lambda \dot{\mathbf{a}}^T \hat{\mathbf{f}} \quad (22)$$

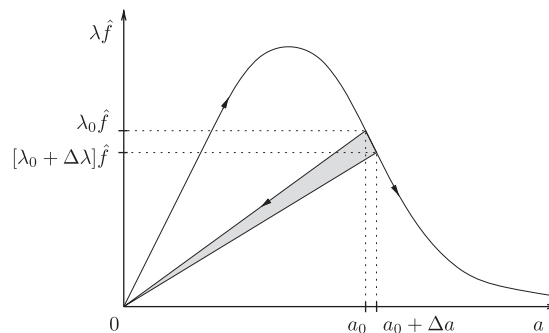


Figure 2. Schematic representation of the dissipation increment (shaded area) in the case of secant unloading. The shaded area is equal to the energy dissipation increment $\tau = \frac{1}{2} \hat{f} (\lambda_0 \Delta a - \Delta \lambda a_0)$, which is a one-dimensional representation of Equation (24).

Substituting this relation and the expression for the exerted power (17) into (16) gives

$$G = \frac{1}{2} \hat{\mathbf{f}}^T (\lambda \dot{\mathbf{a}} - \dot{\lambda} \mathbf{a}) \quad (23)$$

A forward Euler discretization is used to obtain the corresponding incremental path-following constraint as a function of the path parameter τ as, see (11)

$$g = \frac{1}{2} \hat{\mathbf{f}}^T (\lambda_0 \Delta \mathbf{a} - \Delta \lambda \mathbf{a}_0) - \tau \quad (24)$$

Note that λ_0 and $\mathbf{a}_0$ are the converged load factor and the displacements from the previous step, respectively. The derivatives required for the construction of the consistent tangent (13) then read as

$$\frac{\partial g}{\partial \mathbf{a}} = \frac{1}{2} \lambda_0 \hat{\mathbf{f}}^T, \quad \frac{\partial g}{\partial \lambda} = -\frac{1}{2} \hat{\mathbf{f}}^T \mathbf{a}_0 \quad (25)$$

As can be seen the employed forward Euler time discretization yields additional consistent tangent terms that are independent of the displacement increment, making it attractive from an algorithmic point of view. Other time discretizations can be used (e.g. backward Euler), but will lead to more complicated expressions. Moreover, no problems with the forward Euler discretization were encountered in any of the sample problems considered.

In the one-dimensional case, the dissipation increment as formulated in Equation (24) can be illustrated as the shaded area in the force–displacement diagram shown in Figure 2. Note that when the complete equilibrium path is followed, the dissipation increments add up to the maximum energy that can be dissipated by the considered structure (i.e. the total area under the force–displacement diagram).

2.1.2. Geometrically linear model and plasticity. In the case of plasticity, unloading occurs along a path parallel to the elastic tangent (Figure 4). The elastic energy stored in a solid can then be written as

$$V = \frac{1}{2} \int_{\Omega} \boldsymbol{\varepsilon}_e^T \boldsymbol{\sigma} d\Omega \quad (26)$$

where $\boldsymbol{\varepsilon}_e$ is the elastic part of the strain. Since in the case of plasticity the stress $\boldsymbol{\sigma}$ is linearly related to the elastic strain $\boldsymbol{\varepsilon}_e$ via the elastic stiffness matrix $\mathbf{D}_e$, the elastic energy can be rewritten as

$$V = \frac{1}{2} \int_{\Omega} \boldsymbol{\sigma}^T \mathbf{D}_e^{-1} \boldsymbol{\sigma} d\Omega \quad (27)$$

Making use of the symmetry of the elastic stiffness $\mathbf{D}_e$, the rate of elastic energy is derived as

$$\dot{V} = \int_{\Omega} \dot{\boldsymbol{\sigma}}^T \mathbf{D}_e^{-1} \boldsymbol{\sigma} d\Omega = \int_{\Omega} \dot{\boldsymbol{\varepsilon}}^T \mathbf{C}^T \mathbf{D}_e^{-1} \boldsymbol{\sigma} d\Omega \quad (28)$$

with $\mathbf{C} = \partial \boldsymbol{\sigma} / \partial \boldsymbol{\varepsilon}$ being the consistent tangent, which is generally non-symmetric. Using the strain–nodal displacement matrix (19) the rate of elastic energy is obtained as

$$\dot{V} = \dot{\mathbf{a}}^T \mathbf{f}^* \quad (29)$$

where $\mathbf{f}^*$ is a nodal force vector, which is defined as

$$\mathbf{f}^* = \int_{\Omega} \mathbf{B}^T \mathbf{C}^T \mathbf{D}_e^{-1} \boldsymbol{\sigma} d\Omega \quad (30)$$

The energy release rate then follows from Equation (16) as

$$G = \dot{\mathbf{a}}^T (\lambda \hat{\mathbf{f}} - \mathbf{f}^*) \quad (31)$$

where use is made of expression (17) for the exerted power. Note that as long as the deformation of all points in the domain Ω is elastic, the consistent tangent $\mathbf{C}$ is identical to the elastic stiffness $\mathbf{D}_e$. In this case, the nodal force vector $\mathbf{f}^*$ is equal to the internal force vector and the energy release rate G as presented in (31) is equal to zero. Using a forward Euler time discretization the path-following constraint can be expressed as

$$g = \Delta \mathbf{a}^T (\lambda_0 \hat{\mathbf{f}} - \mathbf{f}_0^*) - \tau \quad (32)$$

and the derivatives that are required for the computation of the consistent tangent (13) are obtained as

$$\frac{\partial g}{\partial \mathbf{a}} = (\lambda_0 \hat{\mathbf{f}} - \mathbf{f}_0^*)^T, \quad \frac{\partial g}{\partial \lambda} = 0 \quad (33)$$

Note that, as a consequence of the forward Euler time discretization, the additional force vector $\mathbf{f}^*$ only needs to be computed after each converged load step. This implies that the energy release rate constraint can be applied at the cost of only one more vector assembly per load step. For large systems of equations that make use of the consistent tangent, the additional computational effort will therefore be negligible. An additional advantage of the fact that the additional force vector is only required after each converged state is that it does not depend on the increments of the nodal displacement vector.

In order to indicate the dissipation increment in the case of plasticity in a graphical fashion, consider a one-dimensional bar loaded in tension (Figure 3). Assuming a uniform stress $\sigma = \lambda \hat{f}/A$ and uniform strain $\varepsilon = a/l$, the path parameter yields

$$\tau = \Delta a (\lambda_0 \hat{f} - f_0^*) = \Delta a \left(1 - \frac{C_0}{D_e} \right) \lambda_0 \hat{f} \approx \Delta a_p \lambda_0 \hat{f} \quad (34)$$

where use is made of

$$C_0 \approx \frac{\Delta \sigma}{\Delta \varepsilon} = \frac{l}{A} \frac{\Delta \lambda \hat{f}}{\Delta a} \quad \text{and} \quad D_e \approx \frac{\Delta \sigma}{\Delta \varepsilon_e} = \frac{l}{A} \frac{\Delta \lambda \hat{f}}{\Delta a_e} \quad (35)$$

The rate of dissipation in the case of plasticity (34) is graphically indicated in Figure 4. It needs to be emphasized that the additive decomposition of displacements ($\Delta a = \Delta a_e + \Delta a_p$) is only used for the one-dimensional beam with uniform stress and strain fields. This decomposition is therefore only used for illustrative purposes and not for the derivation of the multi-dimensional constraint (32), which is based on the assumption of the additive decomposition of strains ($\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_e + \boldsymbol{\varepsilon}_p$).

2.1.3. Geometrically non-linear model and damage. In the previous sections, small displacements and strains were assumed. However, in many situations, such assumptions cannot be made. In this

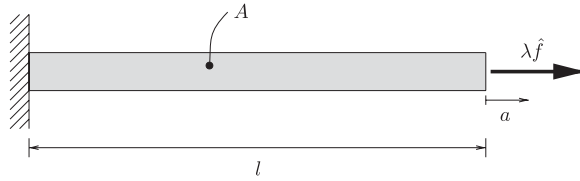


Figure 3. One-dimensional bar loaded in tension.

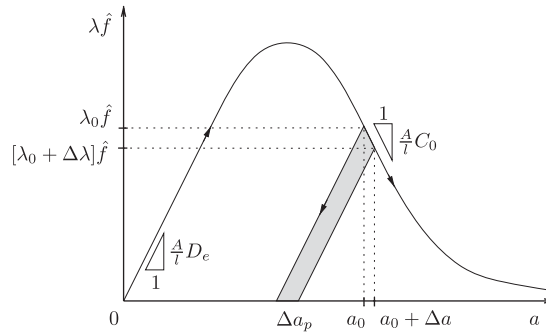


Figure 4. Schematic representation of the dissipation increment (shaded area) in the case of elastic unloading. The shaded area equals $\Delta a_p(\lambda_0 + \frac{1}{2}\Delta\lambda)\hat{f}$. For small increments ($\Delta a, \Delta\lambda \ll 1$), this area is equal to the dissipation increment $\tau = \Delta a(\lambda_0 \hat{f} - f_0^*)$, which is a one-dimensional representation of Equation (32).

section a damage description in combination with a large displacement but small strain formulation is considered. In unloading, a linear elastic relation between the second Piola–Kirchhoff stress and Green–Lagrange strain is assumed.

In such a finite deformation kinematic model, a distinction must be made between the original and the current coordinate system. Using a Lagrangian formulation [9], the internal work can be written as

$$\delta W_{\text{int}} = \int_{\Omega^{t+\Delta t}} \boldsymbol{\sigma}^T \delta \boldsymbol{\varepsilon} d\Omega^{t+\Delta t} \quad (36)$$

where $\Omega^{t+\Delta t}$ is the configuration of a body Ω at time $t + \Delta t$. Generally the integration is not performed over the current configuration but over a reference configuration. In the case of a total Lagrangian formulation, the undeformed configuration (Ω^0) is taken as reference. In the case of an updated Lagrangian formulation, the previously converged solution is taken as reference (Ω^t). Both formulations are suitable for modelling large displacements, large rotations and large strains. The choice of one formulation rather than the other is therefore generally made on the basis of arguments concerning numerical efficiency or ease of implementation. In the case considered here, the total Lagrangian formulation is most suitable since for that formulation the rate of dissipation can easily be expressed in terms of the nodal displacements and forces.

Using the total Lagrangian formulation, the virtual internal work is equal to

$$\delta W_{\text{int}} = \int_{\Omega^0} \mathbf{S}^T \delta \boldsymbol{\gamma} d\Omega^0 \quad (37)$$

where $\boldsymbol{\gamma}$ is the Green–Lagrange strain tensor and $\mathbf{S}$ is the second Piola–Kirchhoff stress, both defined with respect to the initial configuration. The Green–Lagrange strain tensor is defined as

$$\boldsymbol{\gamma} = \frac{1}{2}[(\nabla \mathbf{u} + \mathbf{I})^T(\nabla \mathbf{u} + \mathbf{I}) - \mathbf{I}] = \nabla^s \mathbf{u} + \frac{1}{2}(\nabla \mathbf{u})^T(\nabla \mathbf{u}) \quad (38)$$

An increment of the Green–Lagrange strain can be related to an increment of the nodal displacements via

$$\delta \boldsymbol{\gamma} = \mathcal{B} \delta \mathbf{a} \quad (39)$$

where the $\mathcal{B}$ is the geometrically non-linear equivalent of the $\mathbf{B}$ matrix (6). In contrast to the geometrically linear case, matrix $\mathcal{B}$ depends on the nodal displacement vector: $\mathcal{B} = \mathcal{B}(\mathbf{a})$. Substitution of (39) into (37) then yields the internal force vector

$$\mathbf{f}_{\text{int}} = \int_{\Omega^0} \mathcal{B}^T \mathbf{S} d\Omega^0 \quad (40)$$

Under the assumption of small strains (but large deformations and rotations) it can be assumed that Hooke's law can be applied to relate the second Piola–Kirchhoff stresses to the Green–Lagrange strains in the case of elastic unloading [9]. The internal elastic energy can then be expressed as

$$V = \frac{1}{2} \int_{\Omega^0} \boldsymbol{\gamma}^T \mathbf{S} d\Omega^0 \quad (41)$$

The rate of change of elastic energy can then be derived as

$$\dot{V} = \frac{1}{2} \int_{\Omega^0} \dot{\boldsymbol{\gamma}}^T \mathbf{S} d\Omega^0 + \frac{1}{2} \int_{\Omega^0} \boldsymbol{\gamma}^T \dot{\mathbf{S}} d\Omega^0 \quad (42)$$

Using Equation (39) to substitute the strain rate then yields

$$\dot{V} = \frac{1}{2} \dot{\mathbf{a}}^T \mathbf{f}_{\text{int}} + \frac{1}{2} \int_{\Omega^0} \boldsymbol{\gamma}^T \dot{\mathbf{S}} d\Omega^0 \quad (43)$$

where use is made of (40). The latter term in this expression can be rewritten by using the symmetric material tangent

$$\mathbf{C} = \frac{\partial \mathbf{S}}{\partial \boldsymbol{\gamma}} \quad (44)$$

to yield

$$\dot{V} = \frac{1}{2} \dot{\mathbf{a}}^T \mathbf{f}_{\text{int}} + \frac{1}{2} \int_{\Omega^0} \boldsymbol{\gamma}^T \mathbf{C} \dot{\boldsymbol{\gamma}} d\Omega^0 \quad (45)$$

Using Equations (9) and (39) the rate of elastic energy can be formulated as

$$\dot{V} = \frac{1}{2} \dot{\mathbf{a}}^T (\lambda \hat{\mathbf{f}} + \mathbf{f}^*) \quad (46)$$

in which

$$\mathbf{f}^*(\mathbf{a}) = \int_{\Omega^0} \mathcal{B}^T \mathbf{C} \boldsymbol{\gamma} d\Omega^0 \quad (47)$$

The rate of dissipation can finally be determined using Equation (16) as

$$G = \frac{1}{2} \dot{\mathbf{a}}^T (\lambda \hat{\mathbf{f}} - \mathbf{f}^*) \quad (48)$$

Note that in the case of a linear elastic material, the stress state $\mathbf{S}$ is equal to the symmetrical tangent matrix $\mathbf{C}$ multiplied by the total strain $\boldsymbol{\gamma}$. As a result, the additional force vector $\mathbf{f}^*$ in (47), is equal to the internal force vector $\mathbf{f}_{\text{int}}$ and the dissipated energy G , as formulated in (48), is equal to zero. The path-following constraint is consequently obtained using a forward Euler discretization as

$$g(\Delta \mathbf{a}, \Delta \lambda) = \frac{1}{2} \Delta \mathbf{a}^T (\lambda_0 \hat{\mathbf{f}} - \mathbf{f}_0^*) - \tau \quad (49)$$

The derivatives of the constraint can directly be computed as

$$\frac{\partial g}{\partial \mathbf{a}} = \frac{1}{2} (\lambda_0 \hat{\mathbf{f}} - \mathbf{f}_0^*), \quad \frac{\partial g}{\partial \lambda} = 0 \quad (50)$$

As in the case of geometrically linear damage and geometrically linear plasticity, the area that represents the dissipation increment can also be visualized for the case of geometrically non-linear damage computations. In this case, it is, however, not possible to represent the dissipation increment directly by a simple area (with non-curved edges) in the force–displacement diagram. Alternatively, the dissipation increment can be visualized by an area in the stress–strain curve. In order to reformulate the dissipation increment (49) in terms of the Green–Lagrange strain and second Piola–Kirchhoff stress, the one-dimensional bar loaded in tension (Figure 3) is again considered. In its undeformed state, the bar has length l and cross-sectional area A .

The dissipation increment can in this one-dimensional case be derived from Equation (49) as

$$\tau = \frac{1}{2} \Delta a (\lambda_0 \hat{f} - f_0^*) \quad (51)$$

Since the Green–Lagrange strain γ and corresponding B-matrix (in this one-dimensional case this is actually a scalar) as defined in Equations (38) and (39) can be written as

$$\gamma(a) = \frac{1}{2} \left[\left(\frac{l+a}{l} \right)^2 - 1 \right], \quad \mathcal{B}(a) = \frac{l+a}{l^2} \quad (52)$$

the displacement increment can be approximated by

$$\Delta a \approx \frac{\delta a}{\delta \gamma} \bigg|_{a=a_0} \Delta \gamma = \frac{l^2}{l+a_0} \Delta \gamma \quad (53)$$

The first term in between the parentheses in Equation (51) can be reformulated in terms of the stress and strain by using the internal force vector (40) to yield

$$\lambda_0 \hat{f} = f_{\text{int}}(a_0) = \int_{\Omega^0} \mathcal{B}(a_0) S_0 d\Omega^0 = \frac{l+a_0}{l} A S_0 \quad (54)$$

The second term inside the parentheses in Equation (51) can be obtained from Equation (40) as

$$f_0^* = \int_{\Omega^0} \mathcal{B}(a_0) C_0 \gamma_0 d\Omega^0 = \frac{l+a_0}{l} A C_0 \gamma_0 \quad (55)$$

Substitution of Equations (53), (54) and (55) in the expression for the dissipation increment (51) and dividing by the volume of the bar then give the dissipation increment per unit volume as

$$\frac{\tau}{lA} = \frac{1}{2} [S_0 - C_0 \gamma_0] \Delta \gamma \quad (56)$$

which equals the shaded area in the stress–strain diagram as shown in Figure 5. As can be seen from Figure 5, the derived expression is indeed equal to the dissipation increment.

2.1.4. Integral formulation of the energy release rate constraint. In the three cases considered above, the rate of dissipation can be expressed in terms of nodal quantities. The advantage of having the arc-length constraint defined in terms of nodal quantities is primarily that it is not necessary to evaluate additional volume integrals for the rate of dissipation and its derivatives required for the formulation of the consistent tangent. It should be noted, however, that in the cases of geometrically linear plasticity and geometrically non-linear damage it is anyway necessary to perform an additional vector assembly for the force vectors as presented in Equations (30) and (47). The advantage of expressing the rate of dissipation in terms of nodal quantities is in that case limited.

When considering other types of kinematics or constitutive behaviour, it can be difficult (and in some cases probably impossible) to express the rate of dissipation in terms of nodal quantities. Consider, for example, the case of geometrically non-linear plasticity, where large displacements and rotations but small strains are assumed. By substituting the Green–Lagrange strain and second Piola–Kirchhoff stress for the engineering strain and Cauchy stress in the geometrically linear plasticity model, a geometrically non-linear plasticity formulation is obtained [9]. The elastic

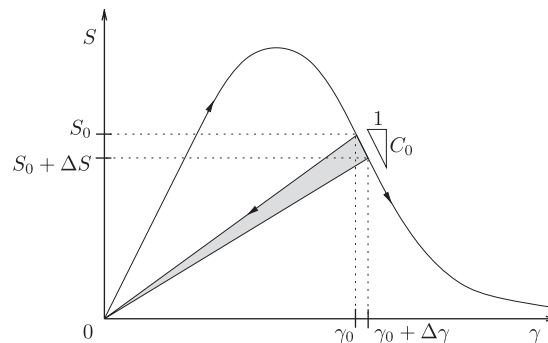


Figure 5. Schematic representation of the dissipation increment per unit volume (shaded area), using a constitutive law in terms of the Green–Lagrange strain γ and second Piola–Kirchhoff stress S . Under the assumption that $\Delta\gamma \ll 1$, the shaded area equals $\frac{1}{2}[S_0 - C_0\gamma_0]\Delta\gamma$, which is equal to the dissipation increment per unit volume as formulated in Equation (56).

energy is then given by

$$V = \frac{1}{2} \int_{\Omega^0} \gamma_e^T \mathbf{S} d\Omega^0 \quad (57)$$

where a total Lagrangian formulation is used and where γ_e is the elastic part of the Green–Lagrange strain, which can be obtained by considering the additive decomposition $\gamma = \gamma_e + \gamma_p$. Rewriting Equation (57) is not straightforward, since the elastic part of the Green–Lagrange strain cannot easily be expressed in terms of nodal quantities.

Alternatively, the derivation of the arc-length constraint in this case can be done by expressing the rate of dissipation (16) as the volume integral

$$G = \int_{\Omega^0} \mathbf{S}^T \left(\dot{\gamma} - \frac{1}{2} \dot{\gamma}_e \right) - \frac{1}{2} \dot{\mathbf{S}}^T \gamma_e d\Omega^0 \quad (58)$$

where the external power (17) has been rewritten as a volume integral. Application of a forward Euler discretization then yields the path-following constraint

$$g = \int_{\Omega^0} \left(\mathbf{S}_0^T \left[\Delta \gamma - \frac{1}{2} \Delta \gamma_e \right] - \frac{1}{2} \Delta \mathbf{S}^T \gamma_{e,0} \right) d\Omega^0 - \tau \quad (59)$$

for which the derivatives required for the construction of the consistent tangent can be derived as

$$\frac{\partial g}{\partial \mathbf{a}} = \int_{\Omega^0} \left(\mathbf{S}_0^T \left[\mathbf{I} - \frac{1}{2} \frac{\partial \gamma_e}{\partial \gamma} \right] - \frac{1}{2} \gamma_{e,0}^T \mathbf{C} \right) d\Omega^0 \mathcal{B}, \quad \frac{\partial g}{\partial \lambda} = 0 \quad (60)$$

with $\mathbf{C}$ being the tangent as defined in (44). Since both the constraint (59) and its derivatives (60) can be determined without much additional computational effort, this integral formulation of the arc-length constraint is an appropriate alternative for the nodal formulation.

3. ALGORITHMIC ASPECTS

The derived path-following constraints are suitable for incorporation in a finite element environment. In order to efficiently and flexibly incorporate the constraints, some algorithmic aspects need further explanation.

3.1. Sherman–Morrison formula

The system of Equations (13) that need to be solved in each Newton–Raphson iteration is an augmented system of the form

$$\begin{bmatrix} \mathbf{K} & -\hat{\mathbf{f}} \\ \mathbf{h}^T & w \end{bmatrix} \begin{bmatrix} d\mathbf{a} \\ d\lambda \end{bmatrix} = \begin{bmatrix} \mathbf{r}^k \\ -g^k \end{bmatrix} \quad (61)$$

where

$$\begin{bmatrix} d\mathbf{a} \\ d\lambda \end{bmatrix} = \begin{bmatrix} \Delta \mathbf{a}^{k+1} \\ \Delta \lambda^{k+1} \end{bmatrix} - \begin{bmatrix} \Delta \mathbf{a}^k \\ \Delta \lambda^k \end{bmatrix} \quad (62)$$

is the incremental solution with respect to the previous iteration k . In finite element computations, matrix $\mathbf{K}$ is in general sparse and banded, which enables a fast solution. The banded structure of the augmented system (61) is destroyed due to the additional column $-\hat{\mathbf{f}}$ and row $\mathbf{h}^T$. As a result, solving the augmented system in a direct manner is not efficient. An alternative approach is to solve the system in parts by applying the Sherman–Morrison formula to yield

$$\begin{bmatrix} d\mathbf{a} \\ d\lambda \end{bmatrix} = \begin{bmatrix} \mathbf{d}^I \\ -g^k \end{bmatrix} - \frac{1}{\mathbf{h}^T \mathbf{d}^I - w} \begin{bmatrix} (\mathbf{h}^T \mathbf{d}^I + g^k) \mathbf{d}^I \\ -\mathbf{h}^T \mathbf{d}^I - g^k (1 + \mathbf{h}^T \mathbf{d}^I - w) \end{bmatrix} \quad (63)$$

where the vectors $\mathbf{d}^I$ and $\mathbf{d}^I$ are obtained by solving the following system of equations:

$$\mathbf{K} \mathbf{d}^I = \mathbf{r}^k, \quad \mathbf{K} \mathbf{d}^I = -\hat{\mathbf{f}} \quad (64)$$

Indeed, the Jacobian of both systems is the same matrix $\mathbf{K}$. As a result, the decomposition of this matrix only needs to be done once per iteration. Since the decomposition is computationally the most intensive part of a solving routine, solving an additional system with the same matrix can efficiently be done.

Note that this approach will only work when the stiffness matrix $\mathbf{K}$ is non-singular. When the stiffness matrix is singular, which may occur when an ideal plastic constitutive relation is used, the complete augmented system must be solved. Furthermore, it should be noted that the set of constraints applied to the augmented system (61) should be translated to constraints for systems (64). The constraints for these systems are derived in Appendix A.

3.2. Prescribed displacements

In the previous sections, all derivations have been carried out on the basis of the assumption that the loading can be described using a scalable external force vector, as presented in Equation (9). Alternatively, a specimen can be loaded by prescribing the displacement of a group of nodes. The nodal displacements can then be decomposed as

$$\mathbf{a} = \mathbf{C} \mathbf{a}_f + \mathbf{a}_p \quad (65)$$

where $\mathbf{a}_f$ and $\mathbf{a}_p$ are, respectively, the free and prescribed nodal displacements. In the case that N_p degrees of freedom are prescribed, the number of free degrees of freedom equals $N_f = N - N_p$. The constraint matrix $\mathbf{C}$ in (65) has size $N \times N_f$. Analogously to the situation of an applied external load $\mathbf{f}_{\text{ext}}$, the prescribed displacement can be written as

$$\mathbf{a}_p = \lambda \hat{\mathbf{a}} \quad (66)$$

where λ is the aforementioned load factor and $\hat{\mathbf{a}}$ is a unit prescribed displacement vector. The equilibrium equation (9) can be reformulated in terms of a residue as

$$\mathbf{r}(\mathbf{a}) = -\mathbf{C}^T \mathbf{f}_{\text{int}}(\mathbf{a}_f, \lambda) = \mathbf{0} \quad (67)$$

Analogously to the derivations in Section 2, a Newton–Raphson scheme can be derived to determine the solution of this equilibrium equation via

$$\begin{bmatrix} \mathbf{C}^T \mathbf{K} \mathbf{C} & \mathbf{C}^T \mathbf{K} \hat{\mathbf{a}} \\ \frac{\partial g}{\partial \mathbf{a}} \mathbf{C} & \frac{\partial g}{\partial \lambda} \hat{\mathbf{a}} \end{bmatrix} \begin{bmatrix} d\mathbf{a}_f \\ d\lambda \end{bmatrix} = \begin{bmatrix} \mathbf{C}^T \mathbf{r}^k \\ -g^k \end{bmatrix} \quad (68)$$

In order to solve this system, the energy release rate constraint and its derivative with respect to the displacement field need to be determined. The required expressions can be derived for the cases considered in Sections 2.1.1–2.1.3.

- In the case of *geometrically linear damage* the rate of elastic energy is given by

$$\dot{V} = \frac{\partial}{\partial t} \left(\frac{1}{2} \mathbf{a}^T \mathbf{f}_{\text{int}} \right) = \frac{1}{2} (\dot{\mathbf{a}}^T \mathbf{f}_{\text{int}} + \mathbf{a}^T \dot{\mathbf{f}}_{\text{int}}) \quad (69)$$

Using Equation (16) in combination with $\mathbf{K}\dot{\mathbf{a}} = \dot{\mathbf{f}}_{\text{int}}$, the energy release rate can be formulated. Employing a forward Euler time discretization yields the path-following constraint

$$g = \frac{1}{2} \Delta \mathbf{a}^T (\mathbf{f}_{\text{int},0} - \mathbf{K}_0^T \mathbf{a}_0) - \tau \quad (70)$$

- The elastic energy rate in the case of *geometrically linear plasticity* is given in Equation (29) as

$$\dot{V} = \dot{\mathbf{a}}^T \mathbf{f}^* \quad (71)$$

with $\mathbf{f}^*$ as defined in (30). The energy release rate is then obtained using (16) and discretized in order to obtain the constraint

$$g = \Delta \mathbf{a}^T (\mathbf{f}_{\text{int},0} - \mathbf{f}_0^*) - \tau \quad (72)$$

- The rate of elastic energy in the case of *geometrically non-linear damage* can be reformulated from Equation (46) as

$$\dot{V} = \frac{1}{2} \dot{\mathbf{a}}^T (\mathbf{f}_{\text{int}} + \mathbf{f}^*) \quad (73)$$

where $\mathbf{f}^*$ is given in Equation (47). The path-following constraint then follows as

$$g = \frac{1}{2} \Delta \mathbf{a}^T (\mathbf{f}_{\text{int},0} - \mathbf{f}_0^*) - \tau \quad (74)$$

With the energy release rate constraints (70), (72) and (74) specified, system (68) can be solved. In Appendix A it is demonstrated that this constrained system can efficiently be solved by application of the Sherman–Morrison formula (63).

3.3. Tracing of non-dissipative parts of an equilibrium path

As mentioned in Section 2.1, robust tracing of the complete equilibrium path requires the path-following parameter to be positive. This requirement leads to problems in the case that non-dissipative parts exist on the equilibrium path. Such parts appear, for example, when a materials initially behaves elastically. Non-dissipative regions on the equilibrium path can, however, also occur in other situations. From a numerical point of view, the use of the energy release rate constraint is also not attractive if the path parameter gets close to the machine precision. Although the rate of dissipation might then be positive, numerical errors can become critical.

For non-dissipative parts of the equilibrium path, alternative path-following constraints must be used. In the case of a geometrically linear computation, a force control (or displacement control) constraint is suitable. In the case of a geometrically non-linear computation, snap-back can occur without any energy being dissipated, making the use of force or displacement control constraints

impossible. In that case, the traditional arc-length constraints [1] can be used to robustly trace the equilibrium path.

The implementation of a robust algorithm for switching from one constraint to another requires two issues to be addressed. The first is the definition of appropriate switching criteria. The second issue is the determination of appropriate step sizes after switching. The treatment of these issues is discussed in some detail in the first numerical experiment.

3.4. Step size adjustment

In order to trace the equilibrium path in as few steps as possible, the path parameter increment needs to be adjusted during the computation. The adjustment is done such that the number of Newton iterations in each loading step is equal to an optimal value k_{opt} . In general, $k_{\text{opt}}=5$ is considered to be an efficient number of iterations in finite element computations for solid mechanics [1]. In [8] it is proposed that the path parameter increment is adjusted according to

$$\tau^i = \tau^{i-1} \frac{k_{\text{opt}}}{k^{i-1}} \quad (75)$$

where $i-1$ refers to the previously converged load step and i refers to the next load step. An alternative criterion can be used as well:

$$\tau^i = \tau^{i-1} \left(\frac{1}{2} \right)^z, \quad z = \frac{k^{i-1} - k_{\text{opt}}}{4} \quad (76)$$

In both cases, the path parameter increment is updated on the basis of its last value only. Experience has shown that a more smooth adjustment of the load increment is beneficial for the robustness of the simulation. This more gradual evolution is obtained by determining the new path parameter increment on the basis of an average path parameter over the history by

$$\tau^i = \frac{1}{n_{\text{his}}} \sum_{j=1}^{n_{\text{his}}} \tau^{i-j} \quad (77)$$

where n_{his} is the history depth. Typical values for this history depth are three or four.

When, for example, plasticity computations are considered, a non-linear system of equations is solved for each integration point. In such a case it is necessary that the local system converges for all integration points, since if it fails for only a single integration point, the global Newton–Raphson iteration will not converge as well. In that case it is therefore required that the magnitude of the dissipation steps is adjusted by the maximum number of local iterations. Typically the local iterations should also converge in five iterations.

4. NUMERICAL EXPERIMENTS

In this section, the performance of the proposed method will be demonstrated by means of five numerical examples. In order to emphasize the versatility of the method, a variety of finite element techniques has been used. In the first example, fracture in a perforated beam is considered. The emphasis here is put on the transition from a force-controlled analysis to the energy constraint analysis and vice versa. In the second example, the method is used in combination with a partition of unity cohesive zone formulation. Next, the energy constraint for plasticity in combination with

a linear kinematic relation will be demonstrated by considering perfect plasticity in an earth slope. As a fourth example, a polycrystal with softening plasticity in its grain boundaries is considered. Finally, the constraint for geometrically non-linear damage will be demonstrated by means of a buckling delamination experiment.

4.1. Fracture of a perforated cantilever beam

An elegant way to simulate crack propagation in brittle or quasi-brittle materials is the cohesive zone method [10–12]. In this method, the process zone, ahead of a crack tip is lumped in a plane. The opening of this plane, the so-called cohesive zone, is governed by an additional constitutive relation. This cohesive constitutive relation, in combination with the balance laws, the constitutive relation for the bulk and the boundary conditions completely specifies the problem.

When the trajectory of the crack is known in advance, e.g. when fracture takes place along well defined interfaces, cohesive zones can be added to the finite element mesh by means of interface elements. These elements consist of two planes (or lines in a two-dimensional simulation), which are connected to the adjacent continuum elements. The relative displacement of the two planes is a measure for the opening $\mathbf{v}$ of the cohesive zone and is used to obtain the traction across the interface $\mathbf{t}=\mathbf{t}(\mathbf{v}, \kappa)$, where κ represents the deformation history of the cohesive constitutive relation. Note that the contribution to the total elastic energy V of these elements is equal to

$$V = \frac{1}{2} \int_{\Omega} \mathbf{v}^T \mathbf{t} d\Omega \quad (78)$$

Further elaboration of this expression yields to the same relations for the path-following constraint g , as given in (24), (32) and (49).

Consider the cantilever beam as shown in Figure 6. The beam is 7.5 mm long and 1 mm thick. The beam is perforated across its entire length by holes with a diameter of 0.2 mm. The spacing of the centre points of these holes is 0.375 mm. The beam is made of a linear elastic material with Young's modulus $E = 100 \text{ N/mm}^2$ and the Poisson ratio $\nu = 0.3$. The ultimate traction for this material is set to $t_{\text{ult}} = 1 \text{ N/mm}^2$ and the fracture toughness is $\mathcal{G}_c = 2.5 \times 10^{-3} \text{ N/mm}$. The beam is loaded by two forces $\lambda \hat{\mathbf{f}}$ as shown in the figure. For the simulation, a plane strain condition is assumed.

Since the geometry of the specimen, its boundary conditions and the applied forces are symmetric with respect to the x -axis, it may be assumed that fracture takes place along this axis. In the

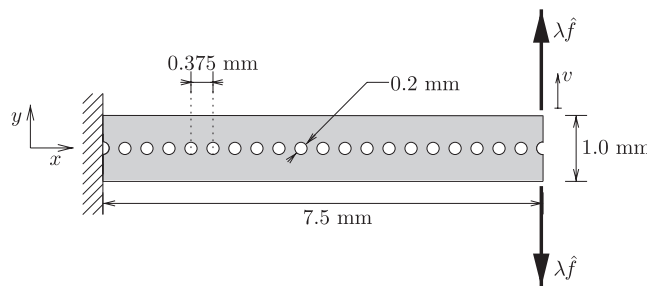


Figure 6. Geometry and loading conditions of a perforated cantilever beam.

finite element model, interface elements have been placed here. Because of symmetry, pure mode-I fracture is assumed as well. This is modelled by a bi-linear damage-based cohesive relation. In order to mimic a perfect bond prior to cracking, a linear dummy stiffness of $D = 1.0 \times 10^4 \text{ N/mm}^3$ is used. Note that because of the linear elastic dummy stiffness, no energy is dissipated in the cohesive zones until the traction reaches the ultimate value t_{ult} .

The considered problem is discretized using 9688 six-node triangles with a seven-point Gauss integration scheme. The pre-defined interface is discretized using 161 six-node interface elements with a three-point Newton–Côtes integration scheme. This discretization results in a total of 40 960 degrees of freedom.

An important aspect in this example is the fact that it is not possible to use the energy constraint method throughout the complete simulation. This is clearly shown in Figure 7, which displays the force–displacement diagram for the perforated beam. Fracture of a segment between two holes will be followed by a part without energy dissipation. In order to robustly trace the equilibrium path, initially a force control constraint is used (indicated by circles $\circ$). When the dissipation increment becomes larger than $1 \times 10^{-8} \text{ Nmm}$, the simulation switches to the energy release constraint (indicated by solid triangles $\blacktriangle$). Although the energy release rate constraint is active, the step size is adjusted by aiming for 5 N iterations per loading step. The maximum allowable dissipation increment τ and load step $\Delta\lambda\hat{f}$ are, respectively, taken as $1 \times 10^{-5} \text{ Nmm}$ and 0.011 N (corresponding to its value in the first load step). The algorithm switches back to a force control constraint when the scaled dissipation increment $\tau/|\Delta\lambda|$ becomes smaller than $1 \times 10^{-10} \text{ Nmm}$. The load steps used for the force-controlled part are taken as the average of the absolute values of the three preceding force increments. During the force-controlled parts of the simulation, the step size is not adjusted. This switching algorithm is also used for the other numerical examples considered in this section.

The shaded area in Figure 7 corresponds to half the amount of energy that is dissipated during the complete fracture of a single segment between two consecutive holes (since the external

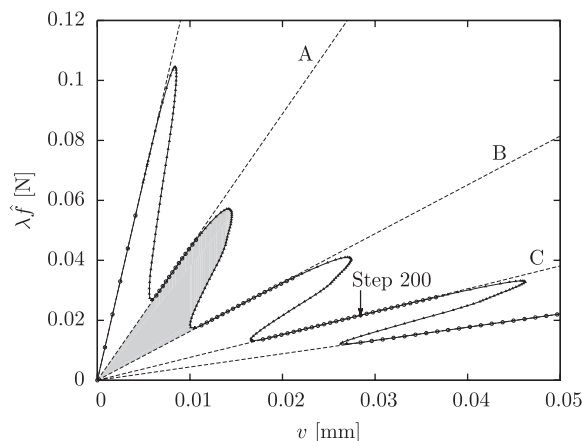


Figure 7. Load–displacement curve of the perforated cantilever beam. Force-controlled steps are indicated by circles ($\circ$) and energy release rate controlled steps by solid triangles ($\blacktriangle$). Note that the shaded area corresponds to half of the energy dissipated during the fracture of a single segment between two consecutive holes. The dashed lines A, B and C represent the elastic load–displacement curves for the cases in which the crack has propagated across 1, 2 and 3 segments, respectively.

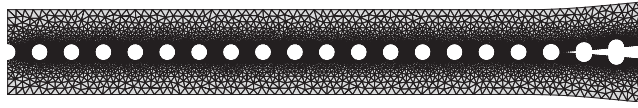


Figure 8. Deformation of the perforated beam (scaled by a factor of 5) after 200 steps.

force is applied at two points), which is equal to the length of these segments times the fracture toughness: $0.175 \times \mathcal{G}_c = 4.375 \times 10^{-4} \text{ Nmm}$. The dashed lines A, B and C represent the elastic load–displacement curves for the cases in which the crack has propagated across 1, 2 and 3 segments, respectively. The deformed specimen is shown in Figure 8.

4.2. Single-edge notched beam using the partition of unity based cohesive zone method

As shown in the previous example, the cohesive zone approach is a robust method to simulate crack propagation. However, an important restriction of this method is that interface elements can only be used when the path of the crack is known beforehand. When the trajectory is not known, alternative formulations can be used. One of these alternatives is to insert the cohesive zone as jumps in the displacement fields of the continuum elements by exploiting the partition-of-unity property of finite element shape functions [13–15]. The magnitude of this displacement jump, which represents the opening of the cohesive zone, is then represented by additional degrees of freedom that are added to the existing nodes of the finite element model.

The partition of unity approach to cohesive fracture has a number of advantages over the conventional models. The cohesive surface can be placed as a discontinuity anywhere in the model, irrespective of the structure of the underlying finite element mesh. Moreover, it is possible to extend a cohesive surface during the simulation. This avoids the use of high dummy stiffnesses to model a perfect bond prior to cracking and prevents numerical problems such as spurious stress oscillations [16] or stress wave reflections in dynamic simulations. Thirdly, since degrees of freedom are only added when a cohesive surface is extended, the total number of degrees of freedom can also be smaller. And finally, the method is based on a cohesive surface formulation that enables the use of existing cohesive constitutive models.

Consider a domain that is crossed by a discontinuity Γ_d that splits the domain into a Ω^+ and a Ω^- part (Figure 9). Here, $\mathbf{n}_d$ is a unit vector normal to the discontinuity pointing into the Ω^+ domain. The discontinuous displacement field $\mathbf{u}$ is constructed by adding a second field $\tilde{\mathbf{u}}$ to the existing displacement field $\hat{\mathbf{u}}$ in the Ω^+ part of the domain only to yield

$$\mathbf{u}(\mathbf{x}, t) = \hat{\mathbf{u}}(\mathbf{x}, t) + \mathcal{H}_{\Gamma_d}(\mathbf{x}) \tilde{\mathbf{u}}(\mathbf{x}, t) \quad (79)$$

where $\mathcal{H}_{\Gamma_d}$ is the Heaviside step function, which is defined as

$$\mathcal{H}_{\Gamma_d} = \begin{cases} 1 & \text{if } \mathbf{x} \in \Omega^+ \\ 0 & \text{if } \mathbf{x} \in \Omega^- \end{cases} \quad (80)$$

This discontinuous displacement field can be used to derive the strain field. The magnitude of the displacement jump $\mathbf{v}$, which represents the opening of the cohesive zone, is equal to the additional displacement field $\tilde{\mathbf{u}}$ at the discontinuity Γ_d .

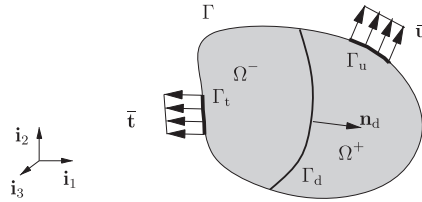


Figure 9. A body Ω that is crossed by a discontinuity Γ_d . Vector $\mathbf{n}_d$ is a unit vector normal to the discontinuity pointing into the Ω^+ domain.

The discontinuous displacement and strain fields can be discretized using standard finite element shape functions that obey the partition of unity. Following Babuška and Melenk [13], the discrete displacement field can be written as

$$\mathbf{u} = \mathbf{N}\mathbf{a} + \mathcal{H}_{\Gamma_d}\mathbf{N}\mathbf{b} \quad (81)$$

where $\mathbf{N}$ is a matrix containing the standard finite element shape functions; $\mathbf{a}$ are the regular degrees of freedom and $\mathbf{b}$ are the enhanced degrees of freedom that construct the displacement field $\tilde{\mathbf{u}}$. The displacement jump at the discontinuity is discretized in a similar manner

$$\mathbf{v} = \mathbf{N}\mathbf{b} \quad (82)$$

The equilibrium equation can now be derived in a similar fashion as discussed in Section 2, Equations (3)–(7), see [15] for details. However, in contrast to the standard situation, two systems of equations are obtained

$$\int_{\Omega} \mathbf{B}^T \boldsymbol{\sigma} d\Omega = \int_{\Gamma_t} \mathbf{N}^T \bar{\mathbf{t}} d\Gamma \quad (83)$$

$$\int_{\Omega} \mathcal{H}_{\Gamma_d} \mathbf{B}^T \boldsymbol{\sigma} d\Omega + \int_{\Gamma_d} \mathbf{N}^T \mathbf{t} d\Gamma = \int_{\Gamma_t} \mathcal{H}_{\Gamma_d} \mathbf{N}^T \bar{\mathbf{t}} d\Gamma \quad (84)$$

where Equation (83), which is identical to (7), is associated with the regular degrees of freedom $\mathbf{a}$ and Equation (84) is associated with the additional degrees of freedom $\mathbf{b}$. In this second equation $\mathbf{t}$ denotes the traction across the discontinuity, which is a function of the displacement jump $\mathbf{v}$.

A discontinuity can be inserted anywhere in the domain by enhancing the nodes that support the elements that are crossed [15]. It is assumed that a discontinuity is straight within a single element and is always extended until it touches the boundary of an element. The propagation of the discontinuity is governed by an additional failure criterion. In this simulation, an equivalent traction algorithm is used [17]. The angle for which the equivalent traction exceeds the maximum allowable traction of the material t_{ult} first is the direction in which the crack propagates. In the model, the discontinuity is extended into the next element until it touches the boundary of that element. An extension of this method, which allows for the nucleation, growth and coalescence of multiple discontinuities is presented in [18].

A limiting factor in the current implementation is the fact that a discontinuity is only extended over a single element after a converged step. As a result, the incremental path-parameter τ is bounded by the amount of energy that dissipates when the crack propagates across an element in

a single load step

$$\tau_{\text{crit}} < \mathcal{G}_c l_e \quad (85)$$

where $\mathcal{G}_c$ is the fracture toughness of the material and l_e is the equivalent length of the corresponding element. Since elements can be crossed in such a manner that the length of the discontinuity in this element is smaller than the equivalent length (and this is especially the case with triangular elements), it is safe to choose a maximum allowable incremental path-parameter that is much smaller than τ_{crit} .

The opening of the cohesive zone is governed by a mixed mode cohesive relation [15]. The normal traction $t_n = \mathbf{t}^T \mathbf{n}_d$ across the discontinuity is equal to

$$t_n = t_{\text{ult}} \exp\left(-\frac{t_{\text{ult}}}{\mathcal{G}_c} v_n\right) \quad (86)$$

where $\mathcal{G}_c$ is the aforementioned fracture toughness of the material, $v_n = \mathbf{v}^T \mathbf{n}_d$ is the normal opening of the discontinuity and t_{ult} represents the strength of the material. The shear traction $t_s = \mathbf{t}^T \mathbf{s}_d$ is defined as ($\mathbf{s}_d$ is a unit vector perpendicular to $\mathbf{n}_d$)

$$t_s = d_{\text{int}} \exp(h_s v_n) v_s \quad (87)$$

where d_{int} is the initial crack shear stiffness at the instant of creation or extension of the segment and $v_s = \mathbf{v}^T \mathbf{s}_d$ is the sliding opening of the discontinuity; h_s defines the softening behaviour in shear direction and is defined as $h_s = \ln(d_{v_n=1.0}/d_{\text{int}})$.

Since the model is based on a small strain formulation and the cohesive relation is of a damage type, the constraint that is used in this example is of the form as described in Section 2.1.1. In the current formulation, both the continuum material and the cohesive zone contribute to the elastic energy. Without losing generality, the total elastic energy V can be written as, cf. (18),

$$V = \frac{1}{2} \int_{\Omega} \boldsymbol{\varepsilon}^T \boldsymbol{\sigma} d\Omega + \frac{1}{2} \int_{\Gamma_d} \mathbf{v}^T \mathbf{t} d\Gamma \quad (88)$$

Using this relation in combination with Equations (83) and (84) gives a constraint function that is identical to the one presented in Equation (24), with this distinction that this equation is defined in the total solution space $[\mathbf{a}, \mathbf{b}]^T$.

The performance of the energy release rate constraint arc-length method in combination with the partition of unity based cohesive zone method is shown in the simulation of a four point bending test of a single-edge notched concrete beam (Figure 10). The specimen is made of concrete with Young's modulus $E = 35.0 \text{ GPa}$ and the Poisson ratio $\nu = 0.15$. The tensile strength and the fracture toughness are $t_{\text{ult}} = 2.8 \text{ MPa}$ and $\mathcal{G}_c = 0.1 \text{ N/mm}$, respectively. The beam has been analysed under plane strain conditions. The mesh consists of 3208 three node triangular elements (Figure 11). In the region in which crack nucleation and propagation is expected, i.e. in the centre of the specimen in between the notch and the supports, the typical length of the element is equal to $l_e = 3 \text{ mm}$.

The simulations are started using a force control method with a unit load of $P = 500 \text{ N}$ and the load factor λ is increased in a linear fashion with increments of $\Delta\lambda = 1.0$. The force control is switched to energy release rate control when the dissipated energy in a single step exceeds the value of $\tau_{\text{init}} = 1 \text{ Nmm}$. The critical path increment is equal to $\tau_{\text{crit}} = 0.1 \times 3.0 \times 100.0 = 30 \text{ Nmm}$ (note that the thickness of the specimen is 100 mm). Taking into account that the mesh is constructed with triangular elements, the maximum allowable path increment is chosen to be $\tau_{\text{max}} = 4 \text{ Nmm}$.

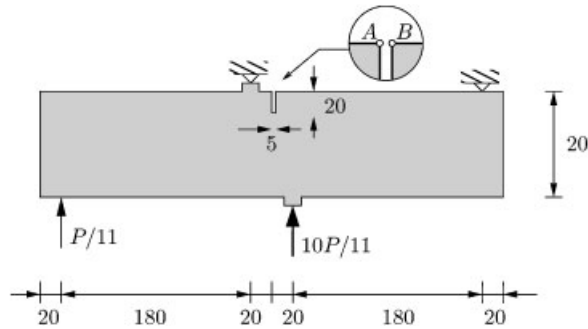


Figure 10. Geometry and loading conditions of the single-edge notched beam. The thickness of the specimen is 100 mm. All dimensions are in millimetres.

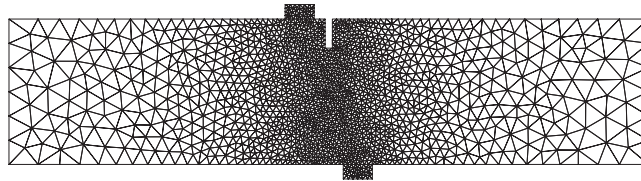


Figure 11. Finite element mesh used for the single-edge notched beam.

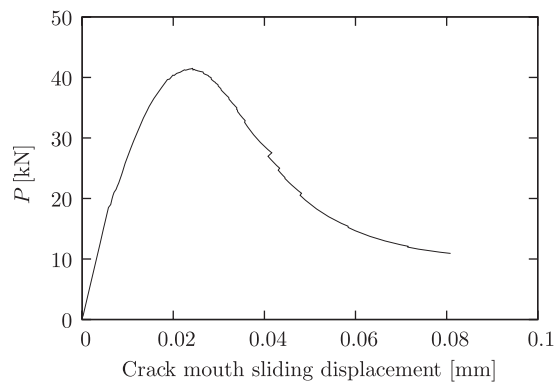


Figure 12. Applied load P versus the crack mouth opening displacement of the single-edge notched beam.

The applied load P versus the crack mouth sliding displacement is given in Figure 12. The crack mouth sliding displacement is defined as the vertical displacement of point A minus the vertical displacement of point B in Figure 10. After load step 51, i.e. when $\lambda=51.0$, the failure criterion is violated at the notch of the specimen and a new cohesive segment is inserted at an angle $\theta=-40.0^\circ$ with the x -axis. The simulation is continued using force control. In load step 77,

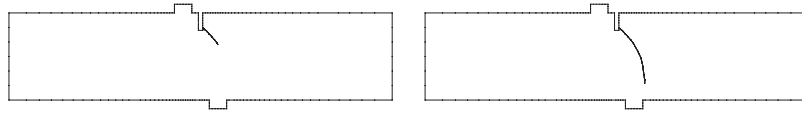


Figure 13. Evolution of the crack in the single-edge notched beam.

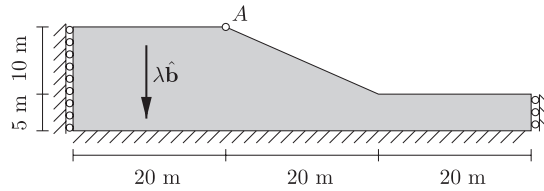


Figure 14. Schematic representation of a slope, loaded by its self-weight.

the dissipated energy exceeds the initialization value $\tau_{\text{init}} = 1.0 \text{ N/mm}$ and the solution procedure is switched to energy release rate control. Since the incremental steps are still relatively small, the system converges in three iterations. As a result, the path parameter increment increases rapidly to the maximum allowable value of $\tau_{\text{max}} = 4.0 \text{ Nmm}$. The maximum load $P = 41.5 \text{ kN}$ is reached after 92 steps.

The evolution of the crack is shown in Figure 13. The position of the crack is in agreement with experimental results [19] and numerical simulations by Rots [20] and Peerlings *et al.* [21]. All these numerical simulations have been performed with a crack mouth opening displacement control.

4.3. Slope stability problem with perfect plasticity

The energy release rate constraint for plasticity (32) is demonstrated using the slope stability problem shown in Figure 14. The considered slope is loaded by increasing its uniform self-weight $\lambda \hat{\mathbf{b}}$ such that the external load vector can be formulated as

$$\mathbf{f}_{\text{ext}} = \lambda \hat{\mathbf{f}} = \lambda \int_{\Omega} \mathbf{N}^T \hat{\mathbf{b}} d\Omega \quad (89)$$

with $\hat{\mathbf{b}}$ being a unit self-weight. Upon increasing the load, the slope will lose stability at a critical value. Since the load cannot be increased after the critical value is attained, a force-controlled simulation is not capable of tracing the complete equilibrium path. A displacement-controlled simulation cannot be used either since the displacement field to be prescribed is a-priori unknown. As can be seen in Figure 15, the plastic zone localizes, which makes the use of an energy release rate constraint attractive for this problem.

The soil slope is modelled as a plane strain elastoplastic continuum. The stresses are given by

$$\dot{\boldsymbol{\sigma}} = \mathbf{D}_e (\dot{\boldsymbol{\varepsilon}} - \dot{\eta} \mathbf{m}) \quad (90)$$

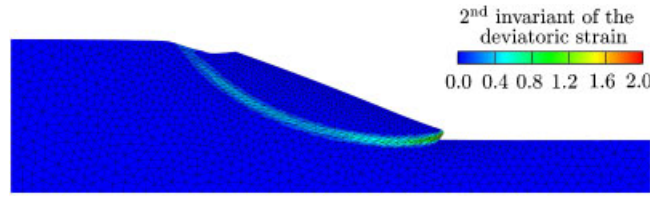


Figure 15. Deformation of the considered slope (scaled by a factor of 25).

where η is the plastic multiplier and $\mathbf{m}$ is the direction of plastic flow. The perfectly plastic Drucker–Prager flow rule

$$f_{\text{yield}} = \sqrt{3J_2} + \frac{1}{3}\alpha I_1 - k = 0 \quad (91)$$

is used, where J_2 and I_1 are the second invariant of the deviatoric stress tensor and the first invariant of the stress tensor, respectively. The Drucker–Prager material constants α and k are related to the internal friction angle ϕ and cohesion c as used for a Mohr–Coulomb flow rule [22] by

$$\alpha = \frac{3 \tan \phi}{\sqrt{3 + 4 \tan^2 \phi}} \quad \text{and} \quad k = \frac{3c}{\sqrt{3 + 4 \tan^2 \phi}} \quad (92)$$

The direction of plastic flow is derived from the plastic potential function

$$\Psi = \sqrt{3J_2} + \frac{1}{3}\beta I_1 + \text{const} \quad (93)$$

as $\mathbf{m} = \partial \Psi / \partial \boldsymbol{\sigma}$, where the coefficient β is related to the dilatancy angle ψ as

$$\beta = \frac{3 \tan \psi}{\sqrt{3 + 4 \tan^2 \psi}} \quad (94)$$

The stress and consistent tangent in each integration point are determined on the basis of the constitutive law (90) in combination with the flow rule (91) and plastic potential function (93) by using a return mapping algorithm [23].

The energy release rate constraint is demonstrated for the problem considered in [24], which is inspired by the slope stability problem in [25]. Young's modulus is taken as 100 MPa and the Poisson ratio is equal to 0.3. The cohesion and internal friction angles are respectively taken as 7.49 kPa and 15.3° and the dilatancy angle is taken as 0°. The slope is discretized using 2854 quadratic (six-node) triangular elements. A three-point Gauss integration scheme is employed in order to avoid locking problems. The considered mesh leads to a total of 11 854 degrees of freedom.

In Figure 16, the self-weight $|\mathbf{b}|$ is plotted versus the downward vertical displacement of point A, v_A , as indicated in Figure 14. The unit self-weight used for this simulation is $\hat{\mathbf{b}} = 1 \text{ kN/m}^3$. As can be seen the slope fails when the self-weight is equal to 20 kN/m^3 . This result differs slightly from the result obtained in [24]. This is most likely a consequence of the use of the Drucker–Prager flow rule instead of the Mohr–Coulomb flow rule. After five force steps, the constraint switches from force control to energy release control. In the simulation shown, the target iteration number is set to 5. As can be seen in Figure 16, the dissipation increment initially increases. Once the

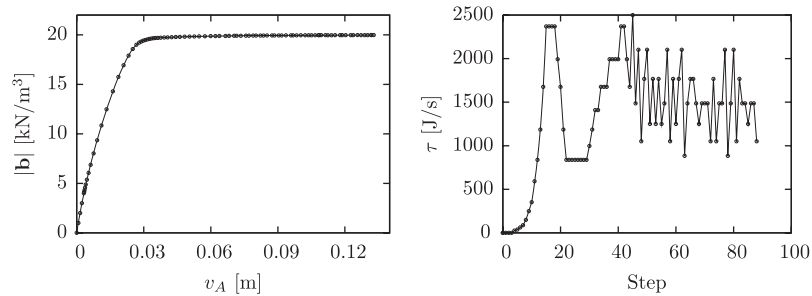


Figure 16. Normalized load scale versus the downward displacement of point A (left) and dissipation increment versus step number (right).

plastic zone starts to develop, over five iteration steps are required and the dissipation steps are adjusted accordingly.

4.4. Polycrystalline structure with interface plasticity

In the previous example, it is demonstrated that the energy release rate constraint (32) for plasticity can also efficiently be used in the case of ideal plasticity. However, softening plasticity is of particular interest since in that case global snap-back can occur.

In order to regularize the problem, the plastic flow is assumed to occur in predefined interfaces [26]. These interfaces are discretized by interface elements that are inserted in the finite element mesh before the computation. In order to determine the contribution of the interface elements to the internal force vector, the traction $\mathbf{t}$ on an interface has to be related to the interface opening $\mathbf{v}$ by

$$\dot{\mathbf{t}} = \mathbf{D}_e(\dot{\mathbf{v}} - \dot{\eta}\mathbf{n}) \quad (95)$$

where η is the plastic multiplier and $\mathbf{n}$ is the normal to the yield surface. In this case the Von Mises yield criterion [27] is used in combination with a work softening (negative hardening) relation. Under the assumption of plane strain, the Von Mises yield criterion can be written as

$$f_{\text{yield}} = \sqrt{3J_2} - \bar{\sigma}(\kappa) = \sqrt{t_n^2 + 3t_t^2} - \bar{\sigma}_0 - h\kappa = 0 \quad (96)$$

where t_n and t_s are, respectively, the normal and tangent components of the traction, $\bar{\sigma}$ is the yield stress, h the hardening parameter and κ the history parameter that evolves according to

$$\dot{\kappa} = \dot{\eta}\mathbf{t}^T\mathbf{n} \quad (97)$$

In each integration point the tractions are computed on the basis of the opening by solving the set of non-linear equations (95) and (96) using a return mapping algorithm [23].

To demonstrate the suitability of this constraint, a polycrystal of 25×25 micrometers (Figure 17) is considered. The grains are generated using a Voronoi algorithm based on a set of points that follow a Poisson distribution [28]. Geometrical periodicity of the polycrystal is enforced in order to apply periodic boundary conditions. The other boundary conditions are shown in Figure 17, where a point load is applied to the bottom-right support in order to deform the polycrystal. The grains are modelled using plane strain continuum elements with modulus of elasticity $E = 200$ GPa

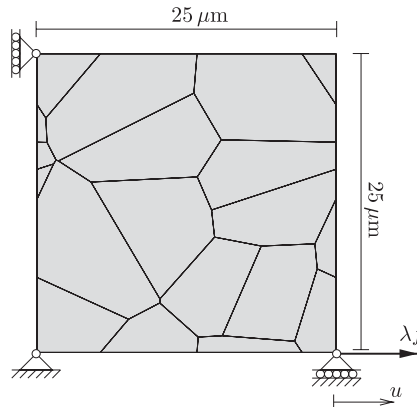


Figure 17. Schematic representation of a geometrically periodic polycrystal generated using a Poisson-Voronoi algorithm, subject to periodic boundary conditions.

and the Poisson ratio $\nu=0.3$. In the interfaces, the yield stress is $\bar{\sigma}_0=200\text{ MPa}$ and two possible choices for the hardening parameter are considered, $h=-55 \times 10^3$ and $-150 \times 10^3 \text{ mm}^{-1}$.

The polycrystal is discretized using 2022 linear triangular elements (i.e. three nodes per element) with Gauss integration scheme. The interfaces are modelled by 190 linear line elements with Newton-Côtes integration scheme. This discretization leads to a total of 2700 degrees of freedom.

In the case that $h=-55 \times 10^3 \text{ mm}^{-1}$ the average horizontal stress over the right edge is plotted versus the average horizontal displacement of that edge. As can be seen in Figure 18, the polycrystal behaves linearly up to 80% of the yield stress, which allows for large load steps. At about 170 MPa some interfaces start to yield and energy is dissipated. The method switches from force control to energy release rate control and the step size is automatically adjusted. Near the maximum load the rate of dissipation is large such that the energy release rate control uses small steps. In the unloading part, initially less energy is dissipated and hence the step size increases again. At some point (displacements larger than $3 \times 10^{-5} \text{ mm}$) the steps get smaller again. In Figure 18 it can be observed that this is caused by the reduced path parameter. The reduction of this parameter is caused by the fact that in this stage the maximum number of return-map iterations is exceeded, due to a highly curved yield surface. Note that the small steps required here are also required in the case that displacement control would be used.

In the case of $h=-150 \times 10^3 \text{ mm}^{-1}$ the snapback is more dramatic than in the case of $h=-55 \times 10^3 \text{ mm}^{-1}$, as can be seen from Figure 19. The path-following constraint is, however, still capable of following the equilibrium path. In Figure 19 it can be seen that the maximum value for the path parameter that is used, is approximately five times smaller than in the previous case. The irregularity in the unloading path is caused by the fact that the yielding initially localizes in a single interface. When this interface failed completely some other interfaces start to yield. The final part of the unloading curve is comparable to that in the case of $h=-55 \times 10^3 \text{ mm}^{-1}$.

4.5. Delamination buckling of a double cantilever beam

The energy release rate constraint for a geometrically non-linear model with damage is demonstrated by considering the double cantilever beam shown in Figure 20. The double cantilever beam

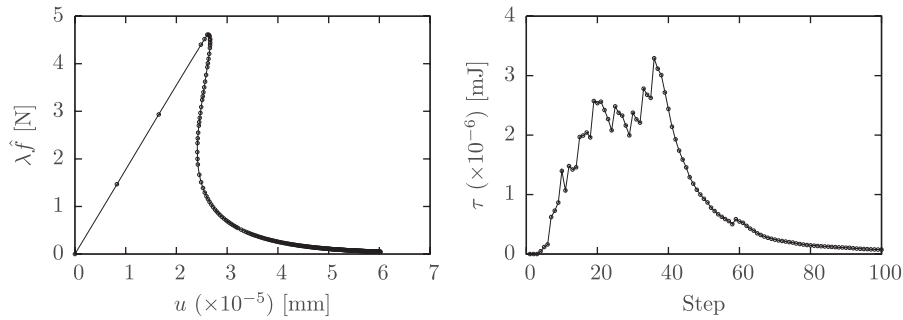


Figure 18. Force–displacement diagram (left) and path parameter increment (right) for the polycrystal with hardening modulus $h = -55 \times 10^3 \text{ mm}^{-1}$.

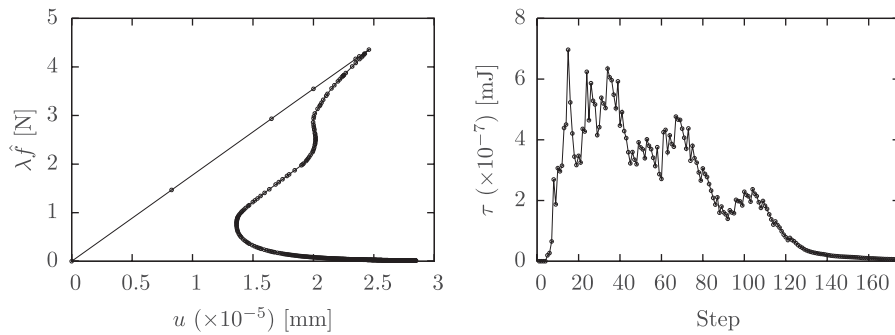


Figure 19. Force–displacement diagram (left) and path parameter increment (right) for the polycrystal with hardening modulus $h = -150 \times 10^3 \text{ mm}^{-1}$.

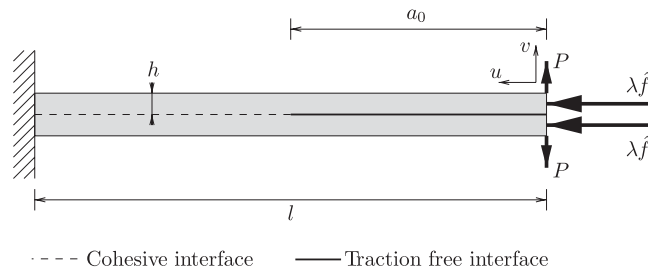


Figure 20. Double cantilever beam with an initial delamination a_0 loaded in compression.

has length $l = 20 \text{ mm}$ and semi height $h = 0.4 \text{ mm}$. An initial delamination of length $a_0 = 10 \text{ mm}$ is present. The beam is loaded by two compressive forces $\lambda \hat{f}$ placed in the midpoints of the upper and lower part and two (small) perturbations P to trigger the correct buckling mode.

Application of interface elements in a geometrically non-linear setting has been discussed in e.g. [29, 30]. Under the assumption that the crack opening remains small as long as the interface has not fully cracked, the elastic energy can be formulated in the undeformed configuration similarly as done in Equation (37) (see [30] for details). As in Equation (88), an additional integral representing the elastic energy in the interface appears. Following the derivation as carried out in Section 2.1.3 yields an expression for the energy release rate with the interface contributions included.

The bulk material is modelled using a plane strain assumption, with modulus of elasticity $E = 135 \text{ GPa}$ and the Poisson ratio $\nu = 0.18$. The interfaces are modelled using interface elements, where the Xu–Needleman [31] constitutive law is used with an ultimate traction $t_{\max} = 75 \text{ MPa}$. Two settings for the fracture toughness are considered, $\mathcal{G}_c = 0.1$ and 0.01 N/mm .

The double cantilever beam is discretized using 1760 quadrilateral elements with 9 nodes per element, which leads to a total of 17 620 degrees of freedom. The cohesive interface is modelled using 400 quadratic interface elements. The unit force $\hat{f}$ is set to 0.1 N and the perturbation forces P are taken as $1.0 \times 10^{-5} \text{ N}$.

The difficulty in this example is that there are two competing non-linear mechanisms. First the part of the specimen that is initially delaminated will buckle locally due to the compressive load. Due to the lateral displacement, the normal interface stress at the tip will increase, which will result in a progressive delamination.

The force–displacement curves for the case that $\mathcal{G}_c = 0.1 \text{ N/mm}$ are shown in Figure 21. As can be seen, the load is increased until the buckling load is reached. At the moment that the interface starts to delaminate, the force drops due to the softening in the interface. This softening continues until the two bars are fully delaminated, after which the force again increases as would be expected from the post-buckling behaviour.

In Figure 21 the upper dashed line indicates the analytical buckling load of a beam with length a_0 which can be computed by using the standard expressions for the critical loads of slender beams loaded in compression

$$F_{\text{cr}}^{a_0} = \frac{\pi^2 E h^3}{48 a_0^2} = 17.77 \text{ N} \quad (98)$$

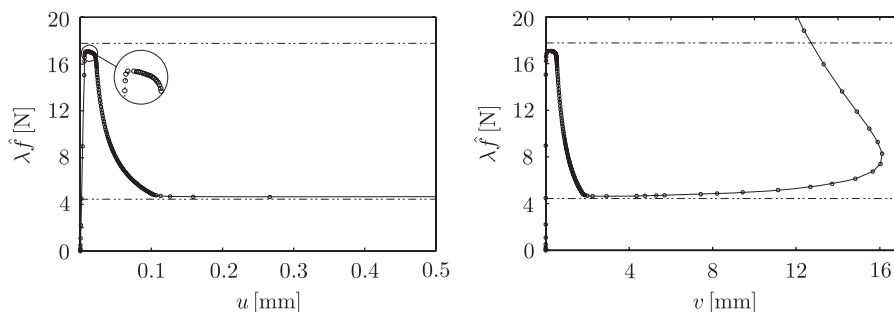


Figure 21. Force–displacement diagrams for the double cantilever beam with fracture toughness $\mathcal{G}_c = 0.1 \text{ N/mm}$. The dashed-dotted lines represent the buckling loads for the original specimen without additional delamination ($F_{\text{cr}} = 17.77 \text{ N}$) and the fully delaminated specimen ($F_{\text{cr}} = 4.44 \text{ N}$).

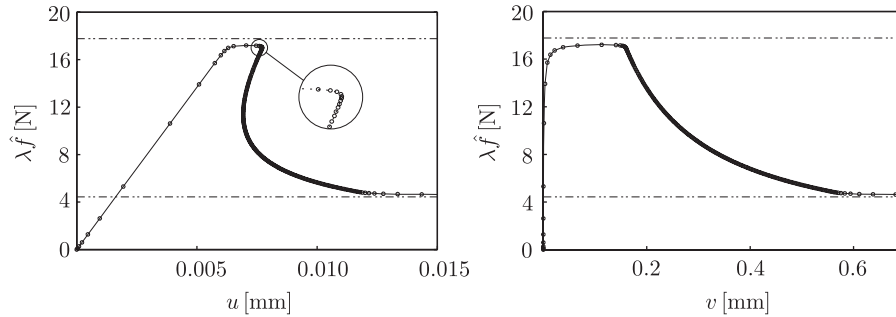


Figure 22. Force–displacement diagrams for the double cantilever beam with fracture toughness $\mathcal{G}_c = 0.01$ N/mm. The dashed-dotted lines represent the buckling loads for the original specimen without additional delamination ($F_{cr} = 17.77$ N) and the fully delaminated specimen ($F_{cr} = 4.44$ N).

and the lower dashed line indicates the critical load in the case of the fully delaminated beam, which equals

$$F_{cr}^l = \frac{\pi^2 E h^3}{48 l^2} = 4.44 \text{ N} \quad (99)$$

As can be seen, both theoretical buckling loads are approached in the computation. The deviations are caused by the fact that the structure is not slender enough in order to neglect the effects of shear forces. This conclusion is supported by the observation that the deviation with respect to the analytical solution is smaller in the case that a beam with length l is considered.

As can be seen from the displacement curve in u -direction, this displacement is monotonically increasing and hence a displacement controlled simulation could be used. This changes, however, in the case that the fracture toughness is lowered to $\mathcal{G}_c = 0.01$ MPa. The force–displacement curves for this case are shown in Figure 22. As can be seen, the horizontal displacement u is now no longer monotonically increasing. As a consequence a displacement controlled computation would fail. The traditional arc length method would also fail, since it would not be able to deal with the localization. Indirect displacement control would still offer a solution in this case, but has the disadvantage that the relevant degrees of freedom need to be selected *a priori*.

5. CONCLUSIONS

A general framework for implementation of an arc-length control on the basis of the energy release rate is proposed. The concept has been applied to the cases of geometrically linear damage, geometrically linear plasticity and geometrically non-linear damage.

It is demonstrated that energy release rate constraints can be derived for the three considered cases as well as for the case of geometrically non-linear plasticity. From a computational point of view, all constraints can be used without much additional computational effort compared to classical arc-length constraints, which makes them attractive alternatives. Although the derivations have been carried out on the assumption of the presence of an external load, it is also demonstrated that it can be applied efficiently in the case of prescribed nodal displacements. In order to robustly

trace non-dissipative parts of the equilibrium path, the used algorithm needs to be capable of switching from force control to energy release rate control and back.

The applicability of the three different energy release rate constraints is shown by means of five numerical examples. The case of geometrically linear damage constraint is demonstrated for two discrete cohesive zone computations. One with pre-defined interface elements and the other based on the partition of unity method. The constraint for geometrically linear plasticity is also demonstrated by means of two numerical examples. The first example is that of a slope stability problem with perfect plasticity. The second example is that of a polycrystal with plastic interfaces. Finally, the geometrically non-linear damage constraint is used to carry out a buckling delamination computation. The examples show that the proposed constraints can be used for robust tracing of the equilibrium path for problems in which material instabilities are involved. This is an important improvement with respect to conventional arc-length methods and derivatives thereof, which are generally less robust in these situations. The implemented switching algorithm is also demonstrated to work well for the problems considered. However, the development of a universally applicable switching algorithm is beyond the scope of this contribution.

It can be concluded that the proposed framework allows for the derivation of arc-length constraints for various types of constitutive behaviour and kinematic relations. Constraints for other cases such as computations with viscous material behaviour have not been derived, but can likely be dealt with in the context of the proposed general framework.

APPENDIX A: APPLICATION OF CONSTRAINTS TO AUGMENTED SYSTEMS SOLVED USING THE SHERMAN–MORRISON FORMULA

The Sherman–Morrison formula (63) gives an efficient way of solving the augmented system (61). In a finite element computation, without application of constraints, matrix $\mathbf{K}$ is in general singular. The application of constraints in combination with the use of the Sherman–Morrison formula is, especially in the case of non-homogeneous constraints, non-trivial.

In order to solve a singular linear system of equations

$$\mathbf{K} \mathbf{d} \mathbf{a} = \mathbf{r} \quad (\text{A1})$$

some of the degrees of freedom $\mathbf{d} \mathbf{a}$ need to be prescribed. This is done using a constraint matrix $\mathbf{C}$ as described in Equation (65) such that

$$\mathbf{d} \mathbf{a} = \mathbf{C} \mathbf{d} \mathbf{a}_f + \mathbf{d} \mathbf{a}_p \quad (\text{A2})$$

with the subscripts ‘f’ and ‘p’, respectively, denoting the free and prescribed degrees of freedom. Note that the expression (A2) is also suitable for application of linear constraints. Hence, the case considered here also covers the case of a prescribed edge. Using Equation (A2), the singular linear system (A1) can be rewritten as

$$[\mathbf{C}^T \mathbf{K} \mathbf{C}] \mathbf{d} \mathbf{a}_f = \mathbf{C}^T (\mathbf{r} - \mathbf{K} \mathbf{d} \mathbf{a}_p) \quad (\text{A3})$$

which can be solved for the free degrees of freedom. In the first iteration, when $k=1$, the prescribed displacements are equal to the total incremental prescribed displacement in the current load step, $\mathbf{d} \mathbf{a}_p = \Delta \mathbf{a}_p$. In the iterations that follow, when $k=2, 3, 4, \dots$, the incremental prescribed

displacements are set to zero, $\mathbf{da}_p = \mathbf{0}$. The solution of the linear system can subsequently be determined by application of Equation (A2).

In the case of the augmented system (61), the constraints equation (A2) can be applied directly to yield the system

$$\begin{bmatrix} \mathbf{C}^T \mathbf{K} \mathbf{C} & -\mathbf{C}^T \hat{\mathbf{f}} \\ \mathbf{h}^T \mathbf{C} & w \end{bmatrix} \begin{bmatrix} \mathbf{da}_f \\ d\lambda \end{bmatrix} = \begin{bmatrix} \mathbf{C}^T (\mathbf{r} - \mathbf{K} \mathbf{da}_p) \\ -g - \mathbf{h}^T \mathbf{da}_p \end{bmatrix} \quad (\text{A4})$$

Note that the augmented system (68) in the case of prescribed displacements as discussed in Section 3.2 is of the same form with $\hat{\mathbf{f}} = -\mathbf{K} \hat{\mathbf{a}}$, $\mathbf{h}^T = \partial g / \partial \mathbf{a}$, $w = (\partial g / \partial \mathbf{a}) \hat{\mathbf{a}}$ and $\mathbf{da}_p = \mathbf{0}$ and hence the following derivations also hold for that case.

The Sherman–Morrison formula can be applied to equation (A4) to solve for $\mathbf{da}_f$ and $d\lambda$ as

$$\begin{bmatrix} \mathbf{da}_f \\ d\lambda \end{bmatrix} = \begin{bmatrix} \mathbf{d}_f^I \\ -g - \mathbf{h}^T \mathbf{da}_p \end{bmatrix} - \frac{1}{\mathbf{h}^T \mathbf{C} \mathbf{d}_f^{\text{II}} - w} \begin{bmatrix} (\mathbf{h}^T [\mathbf{C} \mathbf{d}_f^I + \mathbf{da}_p] + g) \mathbf{d}_f^{\text{II}} \\ -\mathbf{h}^T \mathbf{C} \mathbf{d}_f^I - (g + \mathbf{h}^T \mathbf{da}_p)(1 + \mathbf{h}^T \mathbf{C} \mathbf{d}_f^{\text{II}} - w) \end{bmatrix} \quad (\text{A5})$$

with

$$[\mathbf{C}^T \mathbf{K} \mathbf{C}] \mathbf{d}_f^I = \mathbf{C}^T (\mathbf{r} - \mathbf{K} \mathbf{da}_p) \quad (\text{A6})$$

$$[\mathbf{C}^T \mathbf{K} \mathbf{C}] \mathbf{d}_f^{\text{II}} = -\mathbf{C}^T \hat{\mathbf{f}} \quad (\text{A7})$$

Comparing these two expressions with Equation (A3) shows that these two correspond with the systems

$$\mathbf{K} \mathbf{d}^I = \mathbf{r} \quad (\text{A8})$$

$$\mathbf{K} \mathbf{d}^{\text{II}} = -\hat{\mathbf{f}} \quad (\text{A9})$$

subjected to the constraint

$$\mathbf{d}^I = \mathbf{C} \mathbf{d}_f^I + \mathbf{da}_p \quad (\text{A10})$$

$$\mathbf{d}^{\text{II}} = \mathbf{C} \mathbf{d}_f^{\text{II}} \quad (\text{A11})$$

The solution of the augmented system (61) can be written as

$$\begin{bmatrix} \mathbf{da} \\ d\lambda \end{bmatrix} = \begin{bmatrix} \mathbf{C} \mathbf{da}_f + \mathbf{da}_p \\ d\lambda \end{bmatrix} \quad (\text{A12})$$

Substitution of Equation (A5) and Equations (A8)–(A11) in this equation yields

$$\begin{bmatrix} \mathbf{da} \\ d\lambda \end{bmatrix} = \begin{bmatrix} \mathbf{d}^I \\ -g \end{bmatrix} - \frac{1}{\mathbf{h}^T \mathbf{d}^{\text{II}} - w} \begin{bmatrix} (\mathbf{h}^T \mathbf{d}^I + g) \mathbf{d}^{\text{II}} \\ -\mathbf{h}^T \mathbf{d}^I - g(1 + \mathbf{h}^T \mathbf{d}^{\text{II}} - w) \end{bmatrix} \quad (\text{A13})$$

which equals the Sherman–Morrison formula applied to the unconstrained augmented system (63). This implies that the augmented system (61) with constraints (A2) can be solved using the unconstrained Sherman–Morrison formula (63) under the condition that the Equations (A8) and (A9) are solved with the constraints (A10) and (A11).

Since for solving the linear system (A1) the constraint matrix $\mathbf{C}$ and constraint vector $\mathbf{d}_p$ are also needed, it can be concluded that the constraints can efficiently be applied to the augmented system, while using the Sherman–Morrison formula.

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